

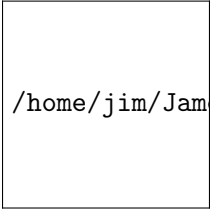
Section 1.4: Analyzing Arguments

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Wednesday 24th September, 2014 & Friday 26th September,
2014



/home/jim/JamesAshe/MHU/images/MarsHillU

Recall: Valid/Invalid Arguments

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Invalid Argument

A deductive argument is *invalid* if the premises are met, but the conclusion does *not necessarily* follow.

- Using a Venn diagram.

Recall: Valid/Invalid Arguments

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Example

If he is president he is a man.

Recall: Valid/Invalid Arguments

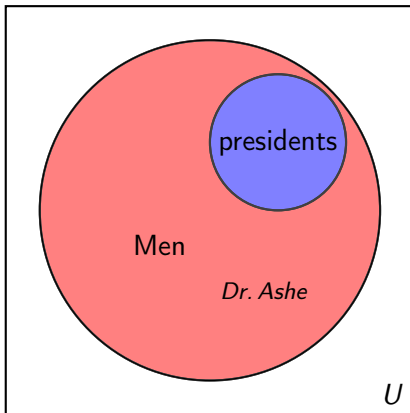
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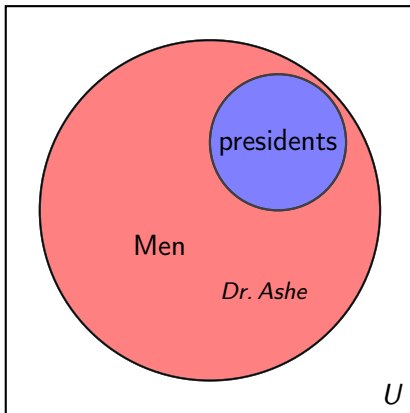
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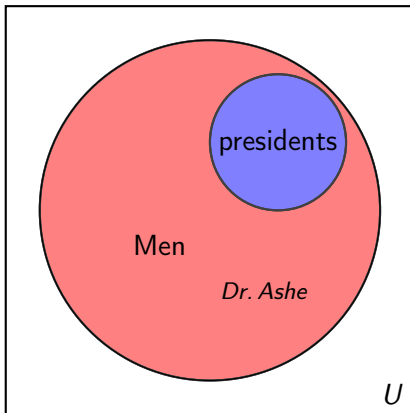
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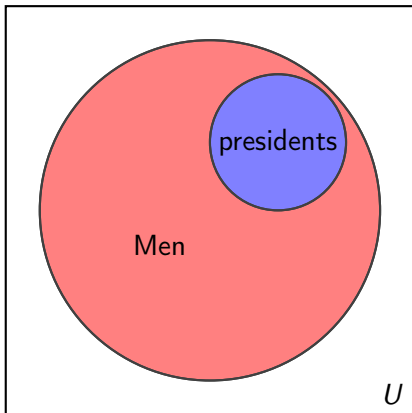
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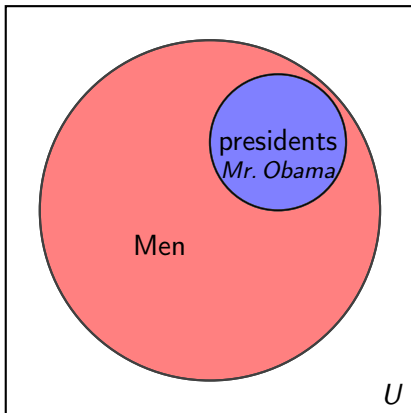
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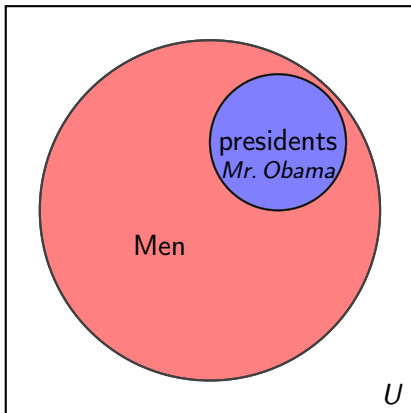
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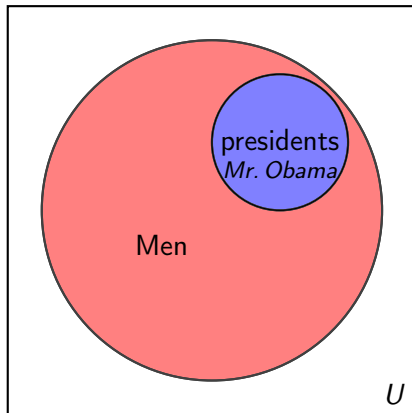
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Analyzing Arguments

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p	m
T	T
T	F
F	T
F	F

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p	m	$p \rightarrow m$	m	$1 \wedge 2$	$c = p$	
T	T					
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- The conclusion is uncertain, so the argument is **invalid**

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p	m					

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p	m					
T	T					
T	F					
F	T					
F	F					

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Hypothesis 1

p	m					
T	T					
T	F					
F	T					
F	F					

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		Hypothesis 1				
p	m	$p \rightarrow q$				
T	T					
T	F					
F	T					
F	F					

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T	F					
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		Hypothesis 1	Hypothesis 2			
p	m	$p \rightarrow q$	p			
T	T					
T	F					
F	T					
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		Hypothesis 1	Hypothesis 2	$H_1 \wedge H_2 \wedge \dots H_n$		
p	m	$p \rightarrow q$	p			
T	T					
T	F					
F	T					
F	F					

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T	F					
F	T					
F	F					

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p	m	$p \rightarrow q$	p	$1 \wedge 2$	$c = m$	
T	T					
T	F					
F	T					
F	F					

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p	m	$p \rightarrow q$	p	$1 \wedge 2$	$c = m$	
T	T					
T	F					
F	T					
F	F					

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p	m	Hypothesis 1 $p \rightarrow q$	Hypothesis 2 p	$H_1 \wedge H_2 \wedge \dots H_n$ $1 \wedge 2$	Conclusion $c = m$	Argument $(1 \wedge 2) \rightarrow c$
T	T					
T	F					
F	T					
F	F					

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T	F	F				
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F	T	T				
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T	F	F	T			
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F	F	T	F			

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T	F	F	T			
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T	T	T	T	T	T	T
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p	m	$p \rightarrow q$	p	$1 \wedge 2$	$c = m$	$(1 \wedge 2) \rightarrow c$
T	T	T	T	T	T	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	F	F	F	T

- The conclusion is certain, so the argument is

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p	m	$p \rightarrow q$	p	$1 \wedge 2$	$c = m$	$(1 \wedge 2) \rightarrow c$
T	T	T	T	T	T	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	F	F	F	T

- The conclusion is certain, so the argument is **valid**

Analyzing Arguments

Analyzing Arguments

- 1 Identify hypotheses and the conclusion.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1.

2.

- 1 Identify hypotheses and the conclusion.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 - 2.
-

- ① Identify hypotheses and the conclusion.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-

- ① Identify hypotheses and the conclusion.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

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Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

- ① Identify hypotheses and the conclusion.
- ② Identify component statements.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

i : The defendant is innocent.
 j : The defendant goes to jail.

- ① Identify hypotheses and the conclusion.
- ② Identify component statements.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

i : The defendant is innocent.
 j : The defendant goes to jail.

- ① Identify hypotheses and the conclusion.
- ② Identify component statements.
- ③ Write in symbolic form.

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Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
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 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

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1.

2.

$\therefore$

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Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

i : The defendant is innocent.
 j : The defendant goes to jail.

1. $i \rightarrow \sim j$

2.

$\therefore$

- ① Identify hypotheses and the conclusion.
- ② Identify component statements.
- ③ Write in symbolic form.

Analyzing Arguments

Problem 1.4.2. Determine whether the following argument is valid

1. If the defendant is innocent,
the defendant does not go to jail.
 2. The defendant does not go to jail.
-
- Therefore the defendant is innocent.

i : The defendant is innocent.
 j : The defendant goes to jail.

1. $i \rightarrow \sim j$
 2. $\sim j$
-
- $\therefore$

- ① Identify hypotheses and the conclusion.
- ② Identify component statements.
- ③ Write in symbolic form.

Analyzing Arguments

- 1 Identify hypotheses and the conclusion.
- 2 Identify component statements.
- 3 Write in symbolic form.

i : The defendant is innocent.
 j : The defendant goes to jail.

$$\begin{array}{l} 1. \quad i \rightarrow \sim j \\ 2. \quad \sim j \\ \hline \therefore i \end{array}$$

Analyzing Arguments

- 1 Identify hypotheses and the conclusion.
- 2 Identify component statements.
- 3 Write in symbolic form.
- 4 Fill out a truth table.

i : The defendant is innocent.
 j : The defendant goes to jail.

$$\begin{array}{l} 1. \quad i \rightarrow \sim j \\ 2. \quad \sim j \\ \hline \therefore \quad i \end{array}$$

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- 1 Identify hypotheses and the conclusion.
- 2 Identify component statements.
- 3 Write in symbolic form.
- 4 Fill out a truth table.

i : The defendant is innocent.
 j : The defendant goes to jail.

$$\begin{array}{l} 1. \quad i \rightarrow \sim j \\ 2. \quad \sim j \\ \hline \therefore i \end{array}$$

		1	2	c	Argument
T	T				
T	F				
F	T				
F	F				

Problem 1.4.3. Use a truth table to determine if the argument is invalid.

If I have a college degree, I am not lazy.

I am lazy.

Therefore, I am lazy.

			1	2		c	Argument
T	T						
T	F						
F	T						
F	F						

Problem 1.4.4. Use a truth table to determine if the argument is invalid.

You do not have financial security if you are a gambler.

You do not have financial security.

Therefore, you are a gambler.

			1	2		c	Argument
T	T						
T	F						
F	T						
F	F						

Problem 1.4.5. Use a truth table to determine if the argument is invalid.

If you argue with a police officer then you get a ticket.

If you do not break the speed limit, you do not get a ticket.

Therefore, if you break the speed limit, you argue with a police officer.

					1	2		C	Arg.
T	T	T							
T	T	F							
T	F	T							
T	F	F							
F	T	T							
F	T	F							
F	F	T							
F	F	F							

Problem 1.4.6. Use a truth table to determine if the argument is invalid.

If you do not recycle, you are not an environmentalist.

If you recycle, you save trees.

Therefore, you are an environmentalist only if you save trees.

						1	2		C	Arg.
T	T	T								
T	T	F								
T	F	T								
T	F	F								
F	T	T								
F	T	F								
F	F	T								
F	F	F								